

Cavity-mediated interactions in a longitudinally-pumped confocal

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We show how the cavity-mediated interactions between atoms placed in a confocal cavity can be understood in terms of the Green's function of an harmonic oscillator. From this, we see clearly that for atoms placed at the cavity center, transverse pattern formation for an atomic BEC can only occur for blue detuned cavity light. Away from the trap center, numerical simulations are required; these show that The connection to Green's functions shows clearly the relation to Talbot pattern formation in free space, where the standard Talbot effect corresponds to a free-space propagator. We discuss the consequences for cavity-induced pairing with fermionic atoms.

I. INTRODUCTION

In the past decade, an area has emerged at the interface of atomic, molecular, optical and condensed matter physics that seeks not to understand the properties of existing matter, but to engineer new states of matter. i.e. rather than asking what theoretical model might explain the behavior seen in a given material, it asks what kinds of behavior are possible within our understanding of physics, and how these behaviors can be realized. The exquisite control available in cold atom systems has been pivotal to many such examples, but despite this, realizing strongly interacting phases has been challenging due to problems of heating and 3-body losses.

A method that has been particularly powerful across a range of physical systems is strong matter-light coupling, creating hybrid quasiparticles, allowing one to tune the dispersion of normal modes. This has been extensively studied for solid state systems coupled to light [?]. For cold atoms, until recently, most experiments have focused on strong coupling to single mode cavities [?], where phenomena such as optomechanical oscillations [?], and superradiant phase transitions [? ? ?] have been studied. While these experiments demonstrate the potential for matter-light coupling to realize paradigmatic models, there are limitations on what kinds of behavior can be realized with single mode cavities. For this reason, some of us have recently developed a multimode (nearly confocal) cavity, in which a BEC of atoms can be trapped so as to realize strong coupling between the cavity modes and the atoms [?].

In the single mode cavity, the cavity can be understood as mediating a long range (all-to-all) interaction between the atoms. As a result, a BEC of atoms in a single mode cavity can be used to realize paradigmatic models of quantum optics, such as the Dicke model [?]. Such models can be realized in various ways, involving either internal degrees of freedom of the atoms [? ?], or the center-of-mass motion of the atoms [? ? ?]. Similarly behavior is also possible for thermal atoms [? ? ?], but such physics is always mean-field-like, due to the all-to-all coupling. Multimode cavities escape this re-

striction. As we discuss further below, they can provide strong cavity mediated interactions with a range controlled by the location of the atoms along the cavity axis, and the detuning of the cavity.

Various aspects of optical and atomic physics in such cavities has already been discussed. In particular, self organization and liquid crystalline order has been predicted for the degenerate cavity subject to a transverse pump beam [? ?], and the effect of the atom cloud on the normal modes of the cavity has been explored [?]. Probing such normal modes involves studying transmission of a weak optical field through the cavity mirrors. If such a probe field is not weak, the possibility arises of self-structuring of the atom cloud, as has been seen experimentally for thermal atoms in a retro-reflected standing wave of light (the Talbot effect) [? ?], and predicted theoretically for a BEC [?]. The aim of this paper is to elucidate the connections between such Talbot physics, cavity mediated interactions, and transverse pattern formation for a BEC in a nearly confocal cavity.

The remainder of this paper is organized as follows

II. MODEL

In this section we derive the equations governing the light field and atoms, for longitudinal pumping in a confocal (or near-confocal) configuration.

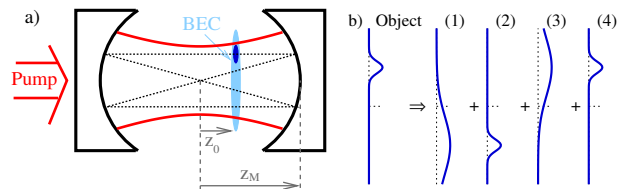


FIG. 1. (a) Cartoon of a confocal cavity, illustrating the beam waist profile (Red), the ray-tracing trajectory (black dashed line), and placement of a pancake of cold atoms (blue). (b) Cavity mediated interactions can be understood as a combination of terms from the four times the ray-tracing profile intersects the cloud of atoms.

As shown in Fig. 1 we consider a cavity formed by two mirrors, of identical curvature R , placed at $z = \pm z_M$. Inside this cavity, there is a cloud of atoms a distance z_0 from the center plane.

A. Modes of confocal cavities

Within the approximation of Gaussian optics[?], we may write the mode functions as $\lambda_{mnq}(\mathbf{r}, z) = u_{mn}(\mathbf{r}, z) \cos(\zeta_{mnq}(z, r))$ labeled by two transverse indices m, n and one longitudinal index q . For now, the transverse mode profile $u_{mn}(\mathbf{r}, z)$ is taken to be a suitably normalized Gauss-Hermite function (see Appendix A). The longitudinal phase then takes the form:

$$\zeta_{mnq}(z, r) = \frac{\omega_{mnq}}{c} \left(z + \frac{r^2}{2R(z)} \right) - (m + n + 1)\psi(z) + \xi_q$$

Here the mode frequency ω_{mnq} must be found by matching boundary conditions at the mirrors, and $\psi(z) = \tan^{-1}(z/z_R)$ is the Guoy phase, written in terms of the Rayleigh range z_R . The Guoy phase term is required to ensure the converging transverse profile is balanced by a change of longitudinal wavevector [?]. The Rayleigh range can be identified from the beam waist $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$ and curvature $R(z) = z + z_R^2/z$. For a symmetric cavity, with mirrors of radius of curvature R_m , one can thus write $z_R = \sqrt{z_M(R_M - z_M)}$.

In order that the modefunction satisfies the boundary conditions we require that $\zeta_{mnq}(z = \mp z_M \pm r^2/R_M, r) = \pi(Q_{\pm} + 1/2)$, where $R_M = R(z)$ is the radius of curvature of the mirrors. Such a condition on ζ_{mnq} means that the cosine term vanishes at both end points. The longitudinal mode-index q counts the difference between the indices $Q_{\pm}$. This allows one to write:

$$\xi_q = \frac{\pi}{2}(q+1), \quad \omega_{mnq} = \frac{c\pi}{4z_M} \left[2q + (m + n + 1) \frac{4\psi(z_M)}{\pi} \right].$$

The behavior simplifies considerably when near confocal geometry, where $z_R = z_M = R_M/2$, and so $4\psi(z_M)/\pi = 1$. This then means that if we choose integer q such that $2q + (m + n + 1) = 2q_0 + 1$, all modes mnq satisfying this equation will have identical frequencies. i.e. this condition defines families of degenerate modes. Since we will consider the case of a high Finesse cavity, we may restrict attention to a single degenerate mode family. Thus we will suppress the q label from hereon, as $q = q_0 - (m + n)/2$. Note also that the argument continues to apply when slightly away from the confocal condition. In this case however $\omega_{mn} = \omega_{00} + (c/z_M)[\psi(z_M) - \pi/4](m + n)$.

Discuss cutoff of high order modes

Note that in the above, one may have q_0 either integer or half integer, corresponding to families of even (odd) parity modes. This restriction (imposed by requiring integer q), means that when summing over mode families, the sum must be restricted to a given parity. This can be

done by inserting a factor $\mathcal{S}_{mn}^{\pm} = (1 \pm (-1)^{m+n})/2$, into the sum for the even(odd) families respectively. Such a factor will be implicitly assumed in the following, by assuming the light field $\lambda(\mathbf{r}, z) = \sum_{mn} \alpha_{mn} \lambda_{mn}(\mathbf{r}, z)$ has non-zero coefficients α_{mn} for only the relevant parity of $m + n$.

B. Atom-mode couplings

Because of the varying Guoy phases of the different cavity modes, it is necessary to consider the interaction

In order to illustrate the transverse pattern formation most cleanly, we consider a two-dimensional pancake of atoms, in the plane orthogonal to the cavity axis (see Fig. 1.) This corresponds to assuming that in the z direction, there is a tight trap, so that the condensate wavefunction can be written in the form $\Phi(\mathbf{r}, z) = \phi(\mathbf{r})Z(z)$. We take the profile $Z(z)$ to be a Gaussian $Z(z) = \exp(-(z - z_0)^2/4\sigma_z^2)/\sqrt{4\pi\sigma_z^2}$, and we assume $\lambda \ll \sigma_z \ll z_R$. As must be the case experimentally, we consider that the atoms experience also a (weaker) in-plane trap, $V_{\text{ext}}(r)$, even before the considering the cavity light.

The system is thus described by two coupled equations, a Schrödinger equation for the transverse atomic wavefunction $\phi(\mathbf{r})$, and an equation for the cavity light field $\lambda(\mathbf{r}, z) = \sum_{mn} \alpha_{mn} \lambda_{mn}(\mathbf{r}, z)$, written in terms of mode coefficients α_{mn} . Writing the former in a frame eliminating the zero-point-energy of the longitudinal profile, and the latter in the rotating frame of the pump laser beam at frequency ω_{Pump} , these coupled equations take the form:

$$i\hbar\partial_t\phi(\mathbf{r}) = -\frac{\hbar^2\nabla^2}{2m}\phi(\mathbf{r}) + V_{\text{ext}}(\mathbf{r})\phi(\mathbf{r}) - E_0 \int dz |Z(z)|^2 |\lambda(\mathbf{r}, z)|^2 \phi(\mathbf{r}), \quad (1)$$

$$i\partial_t\alpha_{mn} = (-\Delta_{mn} - i\kappa)\alpha_{mn} + f_{mn} - E_0 \sum_{m'n'} \int d^2\mathbf{r} |\phi(\mathbf{r})|^2 A_{mn,m'n'}(\mathbf{r}) \alpha_{m'n'}. \quad (2)$$

Here E_0 is the light-shift per atom, i.e. the strength of the AC Stark shift, $E_0 = g_0^2/\delta_a$, where g_0 is the dipole coupling to light, and $\delta_a = \omega_a - \omega_{\text{Pump}}$ the atom-pump detuning, which is assumed large and positive. The source term f_{mn} is the overlap of the pump beam with the mode mn . We have also defined the detuning of the laser from mode mn , i.e. $\Delta_{mn} = \omega_{\text{Pump}} - \omega_{mn}$, so that red detuning corresponds to $\Delta_{mn} < 0$. We have also introduced:

$$A_{mn,m'n'}(\mathbf{r}) = \int dz |Z(z)|^2 u_{mn}^*(\mathbf{r}, z) u_{m'n'}(\mathbf{r}, z).$$

This quantity, corresponding to mode overlaps with the longitudinal atomic density profile, also appears in

Eq. (1). One may readily see that:

$$\int dz |Z(z)|^2 |\lambda(\mathbf{r}, z)|^2 = \sum_{mn m' n'} A_{mn, m' n'}(\mathbf{r}) \alpha_{mn}^* \alpha_{m' n'}. \quad (3)$$

The function $A_{mn, m' n'}(\mathbf{r})$ is thus crucial. It fortunately has a simple form in the limit $\lambda \ll \sigma_z \ll z_R$ which we consider. In this case one finds:

$$A_{mn, m' n'}(\mathbf{r}) = u_{mn}(\mathbf{r}, z_0) u_{m' n'}(\mathbf{r}, z_0) \times \frac{1}{2} \cos \left[\left(\frac{\pi}{4} + \psi(z_0) \right) (m + n - m' - n') \right]. \quad (4)$$

Note that by assuming $\sigma_z \ll z_R$, it is valid to evaluate the transverse mode function at $z = z_0$, as done in the above expression. In the following it will be useful to define $\vartheta_0 = \pi/4 + \psi(z_0)$, and to note that rewriting the cosine as a sum of exponentials allows one to incorporate the effective selection rules imposed by this longitudinal integral via phase factors. We will make particular use of this in the next section.

III. WEAK INTERACTION REGIME – ANALYTIC RESULTS

A. Eliminating cavity modes

In order to gain a simple understanding of the behavior described by Eq. (1,2), we make use of analytic equations which can be derived in the limit of weak matter-light coupling. This means considering the case $E_0 \ll |\omega_{mn} - \omega_{\text{Pump}} - i\kappa|$. Such a limit is tractable as we may solve Eq. (2) perturbatively:

$$\alpha_{mn} = \frac{f_{mn}}{\Delta_{mn} + i\kappa} - E_0 \sum_{m' n'} \int d^2 \mathbf{r} \frac{A_{mn, m' n'}(\mathbf{r}) |\phi(\mathbf{r})|^2 f_{m' n'}}{(\Delta_{mn} + i\kappa)(\Delta_{m' n'} + i\kappa)}.$$

This can then be inserted into Eq. (1), leading to an equation of the form:

$$i\hbar \partial_t \phi(\mathbf{r}) = \left[-\frac{\hbar^2 \nabla^2}{2m} + V_{\text{ext}}(\mathbf{r}) + V_{\text{cav}}(\mathbf{r}) + \int d^2 \mathbf{r}' U_{\text{cav}}(\mathbf{r}, \mathbf{r}') |\phi(\mathbf{r}')|^2 \right] \phi(\mathbf{r}), \quad (5)$$

where $V_{\text{cav}}(\mathbf{r})$ and $U_{\text{cav}}(\mathbf{r}, \mathbf{r}')$ are the pump-induced trap and cavity-mediated interactions respectively.

In the confocal case, where $\Delta_{mn} \equiv \Delta$, we can find closed-form expressions for these terms, as the sum over modes reduces to a sum over the (complete) set of basis functions. This allows one to write all quantities in terms of the two functions:

$$K(\mathbf{r}, \mathbf{r}'; \alpha) = \sum_{mn} S_{mn}^{\pm} u_{mn}(\mathbf{r}, z_0) u_{mn}(\mathbf{r}', z_0) e^{-\alpha(m+n)} \quad (6)$$

$$F(\mathbf{r}, \alpha) = \sum_{mn} S_{mn}^{\pm} f_{mn} u_{mn}(\mathbf{r}, z_0) e^{-\alpha(m+n)}. \quad (7)$$

We have included the symmetry factor explicitly here, as these sums no longer involve the coefficients α_{mn} , which have been eliminated. The coefficient α will allow us to incorporate the phase factors ϑ_0 , as well as to provide a cutoff on the number of transverse modes included.

The function $K(\mathbf{r}, \mathbf{r}'; \alpha)$ has a simple interpretation, which will allow its evaluation below: it is related to propagator for an imaginary time α . The term $(-1)^{m+n}$ in the symmetrization factor can be accounted for by noting that $u_{mn}(-\mathbf{r}) = (-1)^{m+n} u_{mn}(\mathbf{r})$, so that one has:

$$K(\mathbf{r}, \mathbf{r}'; \alpha) = \frac{1}{2} [K_0(\mathbf{r}, \mathbf{r}'; \alpha) \pm K_0(\mathbf{r}, -\mathbf{r}'; \alpha)],$$

where $K_0(\mathbf{r}, \mathbf{r}'; \alpha)$ is the propagator without any symmetrization factor. Explicitly:

$$K_0(\mathbf{r}, \mathbf{r}'; \alpha) = \frac{e^{\alpha}}{w(z_0)^2 \pi \sinh \alpha} \times \exp \left[-\frac{1}{w(z_0)^2} \left(\frac{|\mathbf{r}|^2 + |\mathbf{r}'|^2}{\tanh \alpha} - \frac{2\mathbf{r} \cdot \mathbf{r}'}{\sinh \alpha} \right) \right] \quad (8)$$

As we will see below, the function $F(\mathbf{r}, \alpha)$ can be written in terms of an integral of propagator $K(\mathbf{r}, \mathbf{r}'; \alpha)$ over the pump shape. Before discussing this, let us first write the effective potential and interactions. The potential takes the form:

$$V_{\text{cav}}(\mathbf{r}) = \frac{-E_0}{2(\Delta^2 + \kappa^2)} |F(\mathbf{r}, \alpha_0 - i\vartheta_0)|^2, \quad (9)$$

where α_0 is kept positive to provide a cutoff over high order modes. The expression for the effective interaction is more complex, but as we will discuss, has a simple interpretation, and simple forms in relevant limits. We may write

$$U_{\text{cav}}(\mathbf{r}, \mathbf{r}') = \frac{\Delta E_0^2}{4(\Delta^2 + \kappa^2)^2} \times \Re [F(\mathbf{r}; \alpha_0 - i\vartheta_0) F(\mathbf{r}'; \alpha_0 - i\vartheta_0) K(\mathbf{r}, \mathbf{r}'; \alpha_0 + 2i\vartheta_0) + F(\mathbf{r}; \alpha_0 - i\vartheta_0) F(\mathbf{r}'; \alpha_0 + i\vartheta_0) K(\mathbf{r}, \mathbf{r}'; \alpha_0)], \quad (10)$$

The simple interpretation of the above expression is illustrated in Fig. 1(b); the interaction consists of four terms (incorporating the mirror-images discussed above). These four terms to the parts of the interaction after successive reflections off cavity mirrors. (After four reflections, the pattern repeats for a confocal cavity).

To evaluate the above expressions completely requires also finding an expression for $F(\mathbf{r}, \alpha)$. As noted previously, this can be found by using the propagator $K(\mathbf{r}, \mathbf{r}'; \alpha)$. This follows since the coefficients describing the pump profile can be written as:

$$f_{mn} = \int d^2 \mathbf{r}_M f(\mathbf{r}_M) u(\mathbf{r}_M, z_M),$$

where $f(\mathbf{r}_M)$ is the pump profile on the mirror. The transverse mode profiles at positions z_M and z_0 are

related by $u_{mn}(\mathbf{r}, z_M) = Su_{mn}(S\mathbf{r}, z_0)$ where $S = w(z_0)/w(z_M)$, so that one may write:

$$F(\mathbf{r}, \alpha) = \int d^2\mathbf{r}' f\left(\frac{\mathbf{r}'}{S}\right) K(\mathbf{r}, \mathbf{r}'; \alpha). \quad (11)$$

This then provides a set of closed-form expressions allowing one to evaluate the effective interactions. This is as far as one can proceed in the general case. We next turn to consider various relevant special limits of the above expressions.

B. Simple limits

In most cases we are interested in the behavior for a Gaussian pump spot. Since this is symmetric under $\mathbf{r} \rightarrow -\mathbf{r}$, this means that for an even set of cavity modes we can use $K_0(\mathbf{r}, \mathbf{r}'; \alpha)$ rather than $K(\mathbf{r}, \mathbf{r}'; \alpha)$ in Eq. (11). If the resonant cavity modes are odd, the pump function $F(\mathbf{r}, \alpha)$ vanishes as it is incompatible with an even pump profile. Assuming even modes from hereon, this means a Gaussian profile allows us to replace $F(\mathbf{r}, \alpha) \rightarrow F_G(\mathbf{r}, \alpha)$ where:

$$F_G(\mathbf{r}, \alpha) = \frac{f_0}{\pi w_P^2} \int d^2\mathbf{r}' \exp\left[-\frac{|\mathbf{r}'|^2}{(Sw_P)^2}\right] K_0(\mathbf{r}, \mathbf{r}'; \alpha).$$

This expression may be evaluated as:

$$F_G(\mathbf{r}, \alpha) = \frac{f_0 e^\alpha}{\pi [w(z_M)^2 \sinh \alpha + w_P^2 \cosh \alpha]} \times \exp\left[-\frac{|\mathbf{r}|^2}{w(z_0)^2} \left(\frac{w(z_M)^2 \cosh \alpha + w_P^2 \sinh \alpha}{w(z_M)^2 \sinh \alpha + w_P^2 \cosh \alpha}\right)\right]. \quad (12)$$

As should be obvious, the result is always a Gaussian profile, but it broadens and narrows according to the phase of the complex α . The result simplifies in various limits. For a mode-matched pump spot, $w_P = w(z_M)$ one can readily see that $F_G(\mathbf{r}, \alpha) = (f_0/\pi w_P^2) \exp(-|\mathbf{r}|^2/w(z_0)^2)$, independent of the coefficient α . This means that one may write

$$U_{\text{cav}}(\mathbf{r}, \mathbf{r}') = U_0 \exp\left(-\frac{|\mathbf{r}|^2 + |\mathbf{r}'|^2}{w(z_0)^2}\right) \times \Re[K(\mathbf{r}, \mathbf{r}', \alpha_0 + 2i\vartheta_0) + K(\mathbf{r}, \mathbf{r}', \alpha_0)], \quad (13)$$

where

$$U_0 \equiv \frac{\Delta E_0^2}{4(\Delta^2 + \kappa^2)^2} \left[\frac{f_0}{\pi w_P^2}\right]^2. \quad (14)$$

An even simpler limit exists for a large pump spot size, but requires more work to elucidate. In this case one has

$$F_G(\mathbf{r}, \alpha) = \frac{f_0 e^\alpha}{\pi w_P^2 \cosh \alpha} \exp\left[-\frac{|\mathbf{r}|^2 \tanh \alpha}{w(z_0)^2}\right]. \quad (15)$$

We will consider this limit in combination with the case $\alpha_0 \rightarrow 0$, corresponding to a cavity that confines many

transverse modes. The limit of a small pump spot size is very similar (since a small spot on one mirror will lead to a large spot at the other mirror). We discuss here only the large spot limit, the other limit is simply related.

Because the pump spot, and thus $F(\mathbf{r}, \alpha)$ is symmetric under $\mathbf{r} \rightarrow -\mathbf{r}$, we may write the interaction as: $U_{\text{cav}}(\mathbf{r}, \mathbf{r}') = [U_{\text{cav},0}(\mathbf{r}, \mathbf{r}') + U_{\text{cav},0}(\mathbf{r}, -\mathbf{r}')]/2$ where $U_{\text{cav},0}(\mathbf{r}, \mathbf{r}')$ is defined as in Eq. (10) but with $K(\mathbf{r}, \mathbf{r}'; \alpha)$ replaced by $K_0(\mathbf{r}, \mathbf{r}'; \alpha)$. We may then note the following: In the limit $\alpha \rightarrow 0$, we have

$$K_0(\mathbf{r}, \mathbf{r}'; \alpha \rightarrow 0) = \delta(\mathbf{r} - \mathbf{r}')$$

The prefactor of this term in Eq. (10) becomes $|F_G(\mathbf{r}, i\vartheta_0)|^2$, since the delta function means we only consider the case $\mathbf{r} = \mathbf{r}'$. One may see that $|F_G(\mathbf{r}, i\vartheta_0)|^2 = [f_0/(\pi w_P^2 \cos(\vartheta_0))]^2$. The other contribution to Eq. (10) requires further work. Writing the overall expression as:

$$U_{\text{cav},0}(\mathbf{r}, \mathbf{r}') = \frac{U_0}{\cos^2(\vartheta_0)} [\delta(\mathbf{r} - \mathbf{r}') + u(\mathbf{r}, \mathbf{r}'; \vartheta_0)], \quad (16)$$

where U_0 is given in Eq. (14), we must evaluate the function $u(\mathbf{r}, \mathbf{r}'; \vartheta_0)$ coming from the product $F(\mathbf{r}; \alpha_0 - i\vartheta_0)F(\mathbf{r}'; \alpha_0 - i\vartheta_0)K(\mathbf{r}, \mathbf{r}'; \alpha_0 + 2i\vartheta_0)$. This simplifies considerably using the identity $\tan(\vartheta) + \cot(2\vartheta) = 1/\sin(2\vartheta)$, finally yielding:

$$u(\mathbf{r}, \mathbf{r}'; \vartheta) = \frac{1}{\pi w(z_0)^2 \sin(2\vartheta)} \sin\left[\frac{|\mathbf{r} - \mathbf{r}'|^2}{w(z_0)^2 \sin(2\vartheta)}\right].$$

The above shows that for the limit of large (or small) pump spot size, the interaction takes the form:

$$U_{\text{cav}}(\mathbf{r}, \mathbf{r}') = \frac{U_0}{\cos^2 \vartheta_0} [\delta(\mathbf{r} - \mathbf{r}') + \sigma(\mathbf{r} - \mathbf{r}', \ell(z_0)) + \delta(\mathbf{r} + \mathbf{r}') + \sigma(\mathbf{r} + \mathbf{r}', \ell(z_0))] \quad (17)$$

where $\sigma(\mathbf{s}, \ell) = \sin(|\mathbf{s}|^2/\ell^2)/(\pi\ell^2)$ and $\ell(z)^2 = w(z_0)^2[z_R^2 - z^2]/[z_R^2 + z^2]$. The four terms in Eq. (17) can be directly identified with the the four processes sketched in Fig. 1 (although not in the same order). The lengthscale $\ell(z)$ varies between zero at the cavity ends (in which case $\sigma(\mathbf{s}, \ell \rightarrow 0) = \delta(\mathbf{s})$, and $\ell(0) = w(z_0)^2$, thus the lengthscale for pattern formation varies up to (but not exceeding) the minimum beam waist.

If the cloud density is symmetric, so that $|\phi(-\mathbf{r})|^2 = |\phi(\mathbf{r})|^2$ then we may focus on the first line of Eq. (17). In this case, the momentum space representation is particularly revealing:

$$\begin{aligned} \tilde{U}_{\text{cav}}^{\text{sym}}(\mathbf{q}) &\equiv \int d^2\mathbf{s} e^{i\mathbf{q}\cdot\mathbf{k}} U_{\text{cav}}^{\text{sym}}(\mathbf{r}, \mathbf{r} + \mathbf{s}) \\ &= \frac{2U_0}{\cos^2 \vartheta_0} \left[1 + \cos\left(\frac{q^2 \ell(z_0)^2}{4}\right)\right]. \end{aligned} \quad (18)$$

This form has an immediate consequence: if we consider an uniform cloud, and perform linear stability analysis as discussed in [?], it is clear that for $U_0 > 0$ (blue

detuning), there is no modulational instability, while for $U_0 < 0$ (red detuning), all wavevectors are unstable, and the most unstable wavevectors are $q = \sqrt{2n\pi 2/\ell(z_0)}$, including $q = 0$.

In Refs. [? ? ?], transverse pattern formation was discussed for atoms in a light profile created by a single retroreflected pump beam. In that case, pattern formation was predicted for both red and blue detuning. The relation between the results here, and those in the single mirror case is straightforward. The simplest statement is that whereas our expressions involve Eq. (8), the Green's function of an harmonic oscillator, the equivalent single mirror expressions involve the free space Green's function. As a consequence, reflection off a single mirror leads to an effective interaction $\propto \sigma(\mathbf{r} - \mathbf{r}, \ell)$, where ℓ depends on the distance from the atoms to the mirror. The presence of the cavity adds the other three terms, corresponding to subsequent reflections. The Fourier transform of the term $\sigma(\mathbf{s}, \ell)$ is a cosine term, which can be either positive or negative, hence for either sign of U_0 , pattern formation occurs. In our case, the contributions from all reflections must be included simultaneously, and the contribution of the effective contact interaction cancels the instability for blue detuning.

IV. STRONG INTERACTIONS – NUMERICS

V. DISCUSSION AND CONCLUSIONS

In this paper so far we have focused on bosonic atoms

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Appendix A: Properties of transverse mode functions

In terms of the beam waist $w(z)$, we can write the transverse function as:

$$u_{mn}(\mathbf{r}, z) = \frac{1}{w(z)} \chi_m \left(\frac{\sqrt{2}x}{w(z)} \right) \chi_n \left(\frac{\sqrt{2}y}{w(z)} \right),$$

in terms of the dimensionless profiles

$$\chi_n(x) = \sqrt{\frac{2}{\pi}} \frac{1}{\sqrt{2^n n!}} H_n(x) \exp(-x^2/2),$$

where $H_n(x)$ is the (physicists-)Hermite polynomial.

Appendix B: Longitudinal pattern formation

Equation (1) implicitly assumes a weak cavity field limit, as it ignores atomic recoil in the cavity direction. A relatively simple extension of the equations of motion can be made to include this physics, at least near the concocal point, by writing:

$$\phi(\mathbf{r}, z) = \sum_l \phi_l(\mathbf{r}) \mathcal{N}_l \cos \left(2l \frac{\omega_{00}}{c} z \right) Z(z).$$

where $\mathcal{N}_0 = 1, \mathcal{N}_{l \neq 0} = \sqrt{2}$. This then leads to the coupled Schrodinger equations:

$$\begin{aligned} i\hbar \partial_t \phi_l(\mathbf{r}) = & \left[-\frac{\hbar^2 \nabla^2}{2m} + V_{\text{ext}}(\mathbf{r}) + (2l)^2 \omega_r \right] \phi_l(\mathbf{r}) \\ & - \frac{E_0}{2} \sum_{mn, m'n'} \alpha_{mn}^* u_{mn}(\mathbf{r}, z_0) \alpha_{m'n'} u_{m'n'}(\mathbf{r}, z_0) \\ & \times \left[\mathcal{C}_\Delta \phi_l(\mathbf{r}) + \mathcal{C}_\Sigma \sum_{\delta=\pm 1} \frac{\phi_{l+\delta}(\mathbf{r})}{\mathcal{N}_l \mathcal{N}_{l+\delta}} \right] \end{aligned} \quad (\text{B1})$$

where $\omega_R = k_r^2/2m$, with $k_r = \omega_{00}/2$ being the fundamental wavevector of the cavity modes, and we have written $\mathcal{C}_{\Delta, \Sigma} = \cos[\vartheta_0((m+n) \mp (m'+n'))]$. The corresponding cavity mode equations (using the same notation) become:

$$\begin{aligned} i\partial_t \alpha_{mn} = & (-\Delta_{mn} - i\kappa) \alpha_{mn} + f_{mn} \\ & - \frac{E_0}{2} \sum_{m'n'} \int d^2 \mathbf{r} u_{mn}(\mathbf{r}, z) \alpha_{m'n'} u_{m'n'}(\mathbf{r}, z_0) \\ & \times \sum_l \left[\mathcal{C}_\Delta |\phi_l(\mathbf{r})|^2 + \mathcal{C}_\Sigma \sum_{\delta=\pm 1} \frac{\phi_l^*(\mathbf{r}) \phi_{l+\delta}(\mathbf{r})}{\mathcal{N}_l \mathcal{N}_{l+\delta}} \right]. \end{aligned} \quad (\text{B2})$$

Exploring these full equations allows one to explore longitudinal optomechanical oscillations, as explored in the single-mode cavity [?].